

The ORM in Action

Eloquent Replacing Raw SQL

The GiftList API uses Eloquent, Laravel's ORM. This means we almost never write SQL directly. Instead, we work with PHP objects that represent database records.

The Difference

Here's what fetching a gift looks like without an ORM:

```
// Plain PHP + SQL
$stmt = $pdo->prepare("SELECT * FROM gifts WHERE id = ?");
$stmt->execute([$id]);
$row = $stmt->fetch(PDO::FETCH_ASSOC);

$gift = new Gift();
$gift->id = $row['id'];
$gift->name = $row['name'];
$gift->recipient_id = $row['recipient_id'];
// ... map every field manually
```

And with Eloquent:

```
// Eloquent
$gift = Gift::find($id);
```

One line. The ORM handles the query, the mapping, and the object creation.

Relationships

Eloquent shines with relationships. In GiftList, a Recipient has many Gifts:

```
// In Recipient model
public function gifts()
{
    return $this->hasMany(Gift::class);
}

// Usage
$recipient = Recipient::find($id);
$gifts = $recipient->gifts; // All gifts for this recipient
```

No join queries. No manual mapping. Declare the relationship once; use it everywhere.

Automatic Protection

Every Eloquent query uses parameterized statements. SQL injection is prevented by default:

```
// This is safe-Eloquent escapes the input
$gifts = Gift::where('name', $userInput)->get();
```

You'd have to go out of your way to create a vulnerability.

The 90% Reduction

ORMs can reduce database code by 90% or more. Compare a typical CRUD operation:

Without ORM: Query building, execution, error handling, result fetching, object mapping, relationship loading—easily 50+ lines per operation.

With Eloquent: Model definition plus one-liners for each operation—maybe 10 lines total.

The ORM eliminates boilerplate so you can focus on business logic.

When to Drop Down

Eloquent isn't magic. Complex reports, bulk operations, or performance-critical queries sometimes need raw SQL:

```
$results = DB::select('SELECT ... complex query ...');
```

Good ORMs make this easy. Use the abstraction when it helps; escape it when you need to.